```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/CIP-B107-LAB01/{originals,work}

┌──(kali㉿kali)-[~]
└─$ cd ~/CIP-B107-LAB01
```

```
┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ unzip ~/code.zip -d originals_tmp
Archive:  /home/kali/code.zip
 extracting: originals_tmp/code/check_fine_tune_job_status.py
 extracting: originals_tmp/code/compute_classification_metrics.ipynb
 extracting: originals_tmp/code/create_fine_tune_job.py
 extracting: originals_tmp/code/dataset_small.csv
 extracting: originals_tmp/code/dataset_test.jsonl
 extracting: originals_tmp/code/dataset_train.jsonl
 extracting: originals_tmp/code/file_manager.ipynb
 extracting: originals_tmp/code/fine_tuning_steps.txt
 extracting: originals_tmp/code/predict_ip_single_line.py
 extracting: originals_tmp/code/preprocess_dataset.py
 extracting: originals_tmp/code/result.csv
 extracting: originals_tmp/code/upload_dataset_train.py
 extracting: originals_tmp/code/view_fine_tuning_steps.py

┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ cp originals_tmp/code/{dataset_small.csv,dataset_train.jsonl,dataset_test.jsonl,result.csv,fine_
tuning_steps.txt} originals/

┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ rm -rf originals_tmp
```

```
┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ cp originals_tmp/code/{dataset_small.csv,dataset_train.jsonl,dataset_test.jsonl,result.csv,fine_
tuning_steps.txt} originals/

┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ rm -rf originals_tmp

┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ chmod 444 originals/*

┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ chmod 555 originals

┌──(kali㉿kali)-[~/CIP-B107-LAB01]
└─$ cd originals

┌──(kali㉿kali)-[~/CIP-B107-LAB01/originals]
└─$ sha256sum * | tee ../hashes.txt
74dbffb10b5d9e277f1da6a20d02ccd039b1f6ac754d04c8f621633206ab3e56  dataset_small.csv
77cc78ecec96f5f24175bc894713b54bc5b631b3aaa684665fe2e59663e21e22  dataset_test.jsonl
93938b9d3b627b9dd4126f1d1948e374c10fee5b0733f415572a0fd8b9f35c27  dataset_train.jsonl
0633c4f1029fbbd6e5408f6e251e08eb740544f4a13ebc315ccf3c88e782bd6d  fine_tuning_steps.txt
a2985fb577678515c7ca8dad529beec77e91dd1555b4ed515d43000129e0fe58  result.csv
```

```
┌──(kali㉿kali)-[~/CIP-B107-LAB01/originals]
└─$ stat -c '%n  %s bytes' *
dataset_small.csv  16324 bytes
dataset_test.jsonl  4608 bytes
dataset_train.jsonl  21860 bytes
fine_tuning_steps.txt  31869 bytes
result.csv  41353 bytes

┌──(kali㉿kali)-[~/CIP-B107-LAB01/originals]
└─$ TZ=Africa/Lagos date '+%Y-%m-%d %H:%M:%S %Z (WAT)'; date -u '+%Y-%m-%d %H:%M:%S UTC'
whoami
2026-10-27 14:34:29 WAT (WAT)
2026-10-27 13:34:29 UTC
kali

┌──(kali㉿kali)-[~/CIP-B107-LAB01/originals]
└─$
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python lab01_pipeline.py ../originals
records 229 | columns ['ip', 'class', 'text']
{'A': 177, 'N': 32, 'B': 12, 'C': 4, 'E': 4} | nulls 0 | duplicate rows 77
train 183 {1: 163, 0: 20} | test 46 {1: 34, 0: 12}
strict_valid  False  True
source
negative         24     8
positive          4   193
LEAKAGE: 30/46 test prompts are exact duplicates of training prompts
                 model  TP  FP  FN  TN  accuracy  precision  recall      f1
              majority  34  12   0   0    0.7391     0.7391  1.0000  0.8500
rule (regex+ipaddress)  21   2  13  10    0.6739     0.9130  0.6176  0.7368
  TF-IDF char + LogReg  34   1   0  11    0.9783     0.9714  1.0000  0.9855
ML AUROC 0.9902 | AUPRC 0.9965
              precision    recall  f1-score   support

    negative       1.00      0.92      0.96        12
    positive       0.97      1.00      0.99        34

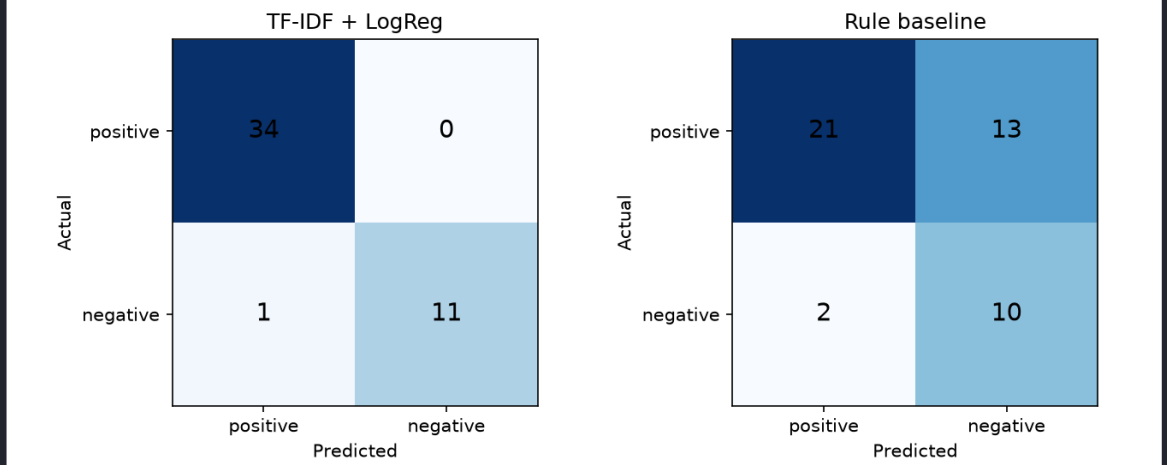
    accuracy                           0.98        46
   macro avg       0.99      0.96      0.97        46
weighted avg       0.98      0.98      0.98        46
```

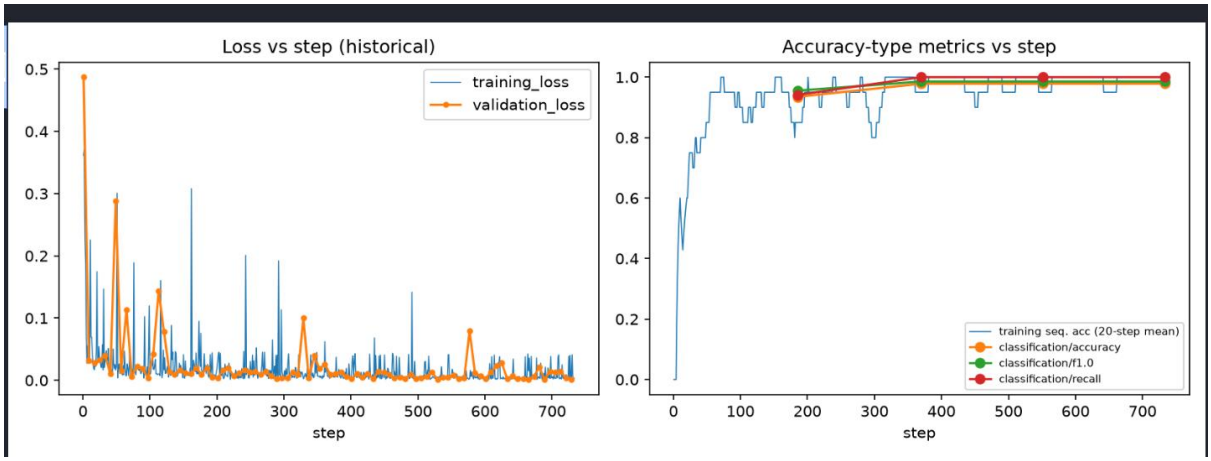
```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python lab01_phaseD_historical.py ../originals
result.csv steps: 733 | fine_tuning_steps.txt steps: 734
non-null counts:
 classification/accuracy          4
classification/precision          4
classification/recall             4
classification/auroc              4
classification/auprc              4
classification/f1.0               4
validation_loss                  92
validation_sequence_accuracy     92
                  metric  final_value  at_step
 classification/accuracy      0.9783       733
classification/precision      0.9714       733
   classification/recall      1.0000       733
    classification/auroc      0.9975       733
    classification/auprc      0.9991       733
     classification/f1.0      0.9855       733

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ ls out/
candidate_review.csv    environment.txt        historical_final_metrics.csv    metrics_summary.csv
confusion_matrices.png  evidence_manifest.csv  historical_training_curves.png  predictions.csv

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$
```






```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ head -n 5 out/candidate_review.csv
source_row,context,candidate,regex_match,ipv4_valid,source_label,analyst_note
2,<string>1.000.000.000+</string>,1.000.000.000,True,False,positive,Source positive but fails strict
 IPv4 (e.g. leading zeros/octet) - preserve disagreement
3,<key>1.0.0.1</key>,1.0.0.1,True,True,positive,
4,<string>1.0.0.1</string>,1.0.0.1,True,True,positive,
5,<key>1.1.1.1</key>,1.1.1.1,True,True,positive,

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 -c "import pandas as pd; r=pd.read_csv('out/candidate_review.csv'); print(r.columns.toli
st()); print('Rows:',len(r))"
['source_row', 'context', 'candidate', 'regex_match', 'ipv4_valid', 'source_label', 'analyst_note']
Rows: 229

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ nano phase_b_evidence.py

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 phase_b_evidence.py
```

```
└─$ python3 phase_b_examples.py

CATEGORY COUNTS
Genuine positive IPv4: 191
Version-like context: 2
Invalid octet above 255: 25
URL-embedded candidate: 26
Valid IPv4 labelled negative: 8
Positive label but invalid IPv4: 4
Numeric prefix/suffix contamination: 45

PHASE B - DIVERSE ACTUAL EVIDENCE EXAMPLES
================================================================================

EXAMPLE 1
Source row: 3
Context: <key>1.0.0.1</key>
Candidate: 1.0.0.1
Regex match: True
Strict IPv4 valid: True
Source label: positive
Review: Valid IPv4 syntax; source label retained.

EXAMPLE 2
Source row: 29
```

```
Context: <string>https://pm-gateway.supersonicads.com/auction?appKey=67052255&amp;SDKVersion=6.8.4.2
&amp;initCountry=US&amp;platform=iphone&amp;advId=0D99ABE9-B1D0-41C1-8C45-2681A498AD97</string>
Candidate: 6.8.4.2
Regex match: True
Strict IPv4 valid: True
Source label: positive
Review: Version-like context; syntactic validity does not establish IP meaning.

EXAMPLE 3
Source row: 7
Context: <string>3608.80.24.1.8</string>
Candidate: 608.80.24.1
Regex match: True
Strict IPv4 valid: False
Source label: negative
Review: At least one IPv4 octet exceeds 255.

EXAMPLE 4
Source row: 48
Context: <string>http://en-us-support.belkin.com/euf/rightnow/js/10.5.0.1.6.137/min/RightNow.js</str
ing>
Candidate: 10.5.0.1
Regex match: True
Strict IPv4 valid: True
```

```
Source label: positive
Review: Candidate is embedded in a longer dotted numeric sequence.

EXAMPLE 5
Source row: 131
Context: <string>0.0.0.0</string>
Candidate: 0.0.0.0
Regex match: True
Strict IPv4 valid: True
Source label: negative
Review: Valid IPv4 syntax but source-labelled negative.

EXAMPLE 6
Source row: 2
Context: <string>1.000.000.000+</string>
Candidate: 1.000.000.000
Regex match: True
Strict IPv4 valid: False
Source label: positive
Review: Candidate fails strict IPv4 validation.

EXAMPLE 7
Source row: 41
Context: <string>135.0.0.22.118 (206072521)</string>
```

```
Candidate: 135.0.0.22
Regex match: True
Strict IPv4 valid: True
Source label: positive
Review: Candidate is embedded in a longer dotted numeric sequence.

EXAMPLE 8
Source row: 5
Context: <key>1.1.1.1</key>
Candidate: 1.1.1.1
Regex match: True
Strict IPv4 valid: True
Source label: positive
Review: Valid IPv4 syntax; source label retained.

EXAMPLE 9
Source row: 8
Context: <string>eng-USA-1.0.0.14-20170731</string>
Candidate: 1.0.0.14
Regex match: True
Strict IPv4 valid: True
Source label: positive
Review: Valid IPv4 syntax; source label retained.
```

```
EXAMPLE 10
Source row: 9
Context: <string>cmn-CHN-1.0.0.12-20170804</string>
Candidate: 1.0.0.12
Regex match: True
Strict IPv4 valid: True
Source label: positive
Review: Valid IPv4 syntax; source label retained.

EXAMPLE 11
Source row: 10
Context: <string>3.6.1.4</string>
Candidate: 3.6.1.4
Regex match: True
Strict IPv4 valid: True
Source label: positive
Review: Valid IPv4 syntax; source label retained.

EXAMPLE 12
Source row: 12
Context: <string>12.6.0.1</string>
Candidate: 12.6.0.1
Regex match: True
Strict IPv4 valid: True
```

```
Source label: positive
Review: Valid IPv4 syntax; source label retained.

EXAMPLES SELECTED: 12
UNIQUE CANDIDATES: 12
UNIQUE SOURCE ROWS: 12
SOURCE RECORDS ONLY: Yes

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ nano phase_b_controlled_tests.py

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 phase_b_controlled_tests.py

PHASE B - CONTROLLED VALIDATION TESTS
================================================================================

Test: Genuine IPv4
Input: 192.168.1.10
Regex candidates: ['192.168.1.10']
Candidate: 192.168.1.10
Strict IPv4 valid: True

Test: Version-like string
```

```
Input: v1.2.3.4
Regex candidates: ['1.2.3.4']
Candidate: 1.2.3.4
Strict IPv4 valid: True

Test: Software version
Input: SDKVersion=6.8.4.2
Regex candidates: ['6.8.4.2']
Candidate: 6.8.4.2
Strict IPv4 valid: True

Test: Invalid octet
Input: 192.168.300.1
Regex candidates: ['192.168.300.1']
Candidate: 192.168.300.1
Strict IPv4 valid: False

Test: URL-embedded IPv4
Input: https://example.test/path/192.168.1.10
Regex candidates: ['192.168.1.10']
Candidate: 192.168.1.10
Strict IPv4 valid: True
```

```
Test: Long dotted sequence
Input: 3608.80.24.1.8
Regex candidates: ['3608.80.24.1']
Candidate: 3608.80.24.1
Strict IPv4 valid: False

CONTROLLED TESTS ONLY - NOT ORIGINAL DATASET RECORDS

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$
```

```
└─$ python3 -c "import pandas as pd; r=pd.read_csv('out/candidate_review.csv'); print(r.head(15).to_
string(index=False)); print('\nColumns:',r.columns.tolist()); print('Total records:',len(r))"
 source_row                                                                                  context      candidate  reg
ex_match  ipv4_valid source_label                                                                                    
          analyst_note                                                                                               
          2                                                                    <string>1.000.000.000+</string>  1.000.000.000
    True       False     positive Source positive but fails strict IPv4 (e.g. leading zeros/octet) -
 preserve disagreement                                                                                               
          3                                                                                   <key>1.0.0.1</key>        1.0.0.1
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
e annotation retained.                                                                                               
          4                                                                                     <string>1.0.0.1</string>        1.0.0.1
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
e annotation retained.                                                                                               
          5                                                                                   <key>1.1.1.1</key>        1.1.1.1
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
e annotation retained.                                                                                               
          6                                                                                     <string>1.1.1.1</string>        1.1.1.1
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
e annotation retained.                                                                                               
          7                                                                    <string>3608.80.24.1.8</string>    608.80.24.1
    True       False     negative                                                                 Candidate fails s
trict IPv4 validation.                                                                                               
          8                                                 <string>eng-USA-1.0.0.14-20170731</string>       1.0.0.14
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
e annotation retained.                                                                                               
          9                                                 <string>cmn-CHN-1.0.0.12-20170804</string>       1.0.0.12
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
e annotation retained.                                                                                               
         10                                                                                     <string>3.6.1.4</string>        3.6.1.4
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
e annotation retained.                                                                                               
         11                                                                    <string>3608.80.24.1.8</string>    608.80.24.1
    True       False     negative                                                                 Candidate fails s
trict IPv4 validation.                                                                                               
         12                                                                                      <string>12.6.0.1</string>       12.6.0.1
    True        True     positive                                  Candidate passes strict IPv4 validation; sourc
```

```
    True        True     positive                    Candidate passes strict IPv4 validation; sourc
e annotation retained.
        14 <string>obfs4 192.95.36.142:443 CDF2E852BF539B82BD10E27E9115A31734E378C2 cert=qUVQ0srL1J
I/vO6V6m/24anYXiJD3QP2HgzUKQtQ7GRqqUvs7P+tG43RtAqdhLOALP7DJQ iat-mode=1</string>  192.95.36.142
    True        True     positive                    Candidate passes strict IPv4 validation; sourc
e annotation retained.
        15    <string>obfs4 38.229.1.78:80 C8CBDB2464FC9804A69531437BCF2BE31FDD2EE4 cert=Hmyfd2ev46
gGY7NoVxA9ngrPF2zCZtzskRTzoWXbxNkzeVnGFPWmrTtILRyqCTjHR+s9dg iat-mode=1</string>    38.229.1.78
    True        True     positive                    Candidate passes strict IPv4 validation; sourc
e annotation retained.
        16   <string>obfs4 38.229.33.83:80 0BAC39417268B96B9F514E7F63FA6FBA1A788955 cert=VwEFpk9F/U
N9JED7XpG1XOjm/O8ZCXK80oPecgWnNDZDv5pdkhq1OpbAH0wNqOT6H6BmRQ iat-mode=1</string>   38.229.33.83
    True        True     positive                    Candidate passes strict IPv4 validation; sourc
e annotation retained.

Columns: ['source_row', 'context', 'candidate', 'regex_match', 'ipv4_valid', 'source_label', 'analys
t_note']
Total records: 229

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ ls -lh out/candidate_review.csv
-rw-rw-r-- 1 kali kali 35K Nov 11 09:07 out/candidate_review.csv

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ head -n 10 out/predictions.csv
prompt,actual,ml_pred_label,ml_score,outcome,rule_pred,majority_pred
       <string>3608.80.24.1.8</string>;608.80.24.1 →,negative,negative,0.11729245642093848,TN,0,1
       <string>2.20.31.4</string>;2.20.31.4 →,positive,positive,0.8065535942630294,TP,1,1
              <string>13.3.1.17D50</string>;13.3.1.17 →,positive,positive,0.7840708668692328,TP,0
,1
              <string>13.3.1.17D50</string>;13.3.1.17 →,positive,positive,0.7840708668692328,TP,0
,1
              <string>104.36.225.166</string>;104.36.225.166 →,positive,positive,0.77108211723917
83,TP,1,1
              <string>104.36.225.154</string>;104.36.225.154 →,positive,positive,0.79137397868613
65,TP,1,1
              <string>104.36.225.179</string>;104.36.225.179 →,positive,positive,0.77517461011023
06,TP,1,1
              <string>104.36.225.142</string>;104.36.225.142 →,positive,positive,0.79481999448111
96,TP,1,1
                     <string>http://en-us-support.belkin.com/euf/rightnow/js/10.5.0.1.6.137/min/R
ightNow.js</string>;10.5.0.1 →,positive,positive,0.727839194480543,TP,0,1

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 -c "import pandas as pd; df=pd.read_csv('out/predictions.csv'); print('Rows:',len(df));
print('Columns:',df.columns.tolist()); print(df.head(10).to_string(index=False))"
Rows: 46
```

```
Columns: ['prompt', 'actual', 'ml_pred_label', 'ml_score', 'outcome', 'rule_pred', 'majority_pred']

                                                                                                  prompt   actual ml_pred_label  ml_score outcome  rule_pred  majority_pred
                                                                                                
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 -c "import pandas as pd; df=pd.read_csv('out/predictions.csv'); print(df.head());"
                                              prompt    actual  ...  rule_pred  majority_pred
0   \t<string>3608.80.24.1.8</string>;608.80.24.1 →  negative  ...          0              1
1         \t<string>2.20.31.4</string>;2.20.31.4 →  positive  ...          1              1
2     \t\t<string>13.3.1.17D50</string>;13.3.1.17 →  positive  ...          0              1
3     \t\t<string>13.3.1.17D50</string>;13.3.1.17 →  positive  ...          0              1
4  \t\t<string>104.36.225.166</string>;104.36.225...  positive  ...          1              1

[5 rows x 7 columns]

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 -c "import pandas as pd; df=pd.read_csv('out/predictions.csv'); print('OUTCOME COUNTS');
 print(df['outcome'].value_counts()); print('\nACTUAL LABELS'); print(df['actual'].value_counts());
print('\nMODEL PREDICTIONS'); print(df['ml_pred_label'].value_counts())"
OUTCOME COUNTS
outcome
TP    34
TN    11
FP     1
Name: count, dtype: int64

ACTUAL LABELS
actual
positive    34
negative    12
Name: count, dtype: int64

MODEL PREDICTIONS
ml_pred_label
positive    35
negative    11
Name: count, dtype: int64

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 -c "import pandas as pd; df=pd.read_csv('out/predictions.csv'); errors=df[df['outcome'].
isin(['FP','FN'])]; print(errors.to_string(index=True))"
                                                                                prompt    actual ml_pred_lab
el  ml_score outcome  rule_pred  majority_pred
21  \t\t\t<string>tincan_register_device_256.0.1.26.113</string>;256.0.1.26 →  negative      positi
ve  0.676402      FP          0              1

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 -c "import pandas as pd; df=pd.read_csv('out/predictions.csv'); print(df[df['outcome']==
'TN'].to_string(index=True))"
                                                                                prompt    actual ml_pred_lab
el  ml_score outcome  rule_pred  majority_pred
0                           \t<string>3608.80.24.1.8</string>;608.80.24.1 →  negative      negati
ve  0.117292      TN          0              1
19                           \t<string>257.0.1.38.116</string>;257.0.1.38 →  negative      negati
ve  0.356960      TN          0              1
20                           \t<string>257.0.1.38.116</string>;257.0.1.38 →  negative      negati
ve  0.356960      TN          0              1
22  \t\t\t<string>tincan_register_device_257.0.1.38.116</string>;257.0.1.38 →  negative      negati
ve  0.384676      TN          0              1
23                           \t<string>257.0.1.38.116</string>;257.0.1.38 →  negative      negati
ve  0.356960      TN          0              1
24                           \t<string>257.0.1.38.116</string>;257.0.1.38 →  negative      negati
ve  0.356960      TN          0              1
34                          \t<string>3608.80.24.1.8</string>;608.80.24.1 →  negative      negati
ve  0.117292      TN          0              1
37                          \t<string>3608.80.24.1.8</string>;608.80.24.1 →  negative      negati
ve  0.117292      TN          0              1
41                          \t<string>3608.80.24.1.8</string>;608.80.24.1 →  negative      negati
ve  0.117292      TN          0              1
43                           \t\t\t\t\t\t<string>0.0.0.0</string>;0.0.0.0 →  negative      negati
ve  0.264877      TN          1              1
45                           \t\t\t\t\t\t<string>0.0.0.0</string>;0.0.0.0 →  negative      negati
ve  0.264877      TN          1              1

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 -c "import pandas as pd; df=pd.read_csv('out/predictions.csv'); print(df[df['outcome']==
```

```
└─$ python3 -c "import pandas as pd; df=pd.read_csv('out/predictions.csv'); print(df[df['outcome']==
'TP'].head(15).to_string(index=True))"

          prompt    actual ml_pred_label  ml_score outcome  rule_pred  majority_pred
1                                                                        \t<string>2.20.31.4</str
ing>;2.20.31.4 →  positive      positive  0.806554      TP          1              1
2                                                                      \t\t<string>13.3.1.17D50</str
ing>;13.3.1.17 →  positive      positive  0.784071      TP          0              1
3                                                                        \t\t<string>13.3.1.17D50</str
ing>;13.3.1.17 →  positive      positive  0.784071      TP          0              1
4                                                                  \t\t<string>104.36.225.166</string>;
104.36.225.166 →  positive      positive  0.771082      TP          1              1
5                                                                  \t\t<string>104.36.225.154</string>;
104.36.225.154 →  positive      positive  0.791374      TP          1              1
6                                                                  \t\t<string>104.36.225.179</string>;
104.36.225.179 →  positive      positive  0.775175      TP          1              1
7                                                                  \t\t<string>104.36.225.142</string>;
104.36.225.142 →  positive      positive  0.794820      TP          1              1
8   \t\t\t<string>http://en-us-support.belkin.com/euf/rightnow/js/10.5.0.1.6.137/min/RightNow.js</st
ring>;10.5.0.1 →  positive      positive  0.727839      TP          0              1
9                                                                        \t<string>20.10.0.8</str
ing>;20.10.0.8 →  positive      positive  0.578453      TP          1              1
10                                                                     \t<string>135.0.0.22.118</stri
ng>;135.0.0.22 →  positive      positive  0.769888      TP          0              1
11                                                                 \t\t<string>34.227.164.166</string>;
34.227.164.166 →  positive      positive  0.748444      TP          1              1
12  \t\t\t<string>http://en-us-support.belkin.com/euf/rightnow/js/10.5.0.1.6.137/min/RightNow.js</st
ring>;10.5.0.1 →  positive      positive  0.727839      TP          0              1
13                                                  \t\t<string>eng-USA-OfflineIOS-7.0.0.4-20200319</s
tring>;7.0.0.4 →  positive      positive  0.825261      TP          1              1
14                                                                           \t<string>7.4.1.2</s
tring>;7.4.1.2 →  positive      positive  0.690181      TP          1              1
15                                                                     \t<string>64.98.122.70</string
>;64.98.122.70 →  positive      positive  0.630641      TP          1              1

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 phase_e_inspect.py

PREDICTION DATASET SUMMARY
Total prediction rows: 46
Unique evidence/prediction combinations: 25
Repeated rows: 21

UNIQUE RECORDS FOR ANALYST REVIEW
================================================================================

Prediction index: 0
Source row(s): 7, 11, 43, 44, 173, 182, 185, 219, 222, 226
Candidate: 608.80.24.1
Context: <string>3608.80.24.1.8</string>
Actual label: negative
Model prediction: negative
Outcome: TN
Model score: 0.11729245642093848
Repeated prediction rows: 4

Prediction index: 1
Source row(s): 127, 142, 149, 174, 179, 186, 211, 225
Candidate: 2.20.31.4
```

```
Context: <string>2.20.31.4</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.8065535942630294
Repeated prediction rows: 3

Prediction index: 2
Source row(s): 187, 188
Candidate: 13.3.1.17
Context: <string>13.3.1.17D50</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7840708668692328
Repeated prediction rows: 2

Prediction index: 4
Source row(s): 189
Candidate: 104.36.225.166
Context: <string>104.36.225.166</string>
Actual label: positive
Model prediction: positive
Outcome: TP
```

```
Model score: 0.7710821172391783
Repeated prediction rows: 1

Prediction index: 5
Source row(s): 190
Candidate: 104.36.225.154
Context: <string>104.36.225.154</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7913739786861365
Repeated prediction rows: 1

Prediction index: 6
Source row(s): 191
Candidate: 104.36.225.179
Context: <string>104.36.225.179</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7751746101102306
Repeated prediction rows: 1

Prediction index: 7
```

```
Source row(s): 192
Candidate: 104.36.225.142
Context: <string>104.36.225.142</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7948199944811196
Repeated prediction rows: 1

Prediction index: 8
Source row(s): 48, 125, 129, 172, 183, 193, 197, 213, 215, 218, 224
Candidate: 10.5.0.1
Context: <string>http://en-us-support.belkin.com/euf/rightnow/js/10.5.0.1.6.137/min/RightNow.js</str
ing>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.727839194480543
Repeated prediction rows: 6

Prediction index: 9
Source row(s): 148, 194, 201
Candidate: 20.10.0.8
Context: <string>20.10.0.8</string>
```

```
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.578453366384503
Repeated prediction rows: 2

Prediction index: 10
Source row(s): 45, 141, 195
Candidate: 135.0.0.22
Context: <string>135.0.0.22.118</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.769887757447613
Repeated prediction rows: 1

Prediction index: 11
Source row(s): 196
Candidate: 34.227.164.166
Context: <string>34.227.164.166</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7484435702178677
```

```
Repeated prediction rows: 1

Prediction index: 13
Source row(s): 89, 198
Candidate: 7.0.0.4
Context: <string>eng-USA-OfflineIOS-7.0.0.4-20200319</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.8252606522195567
Repeated prediction rows: 1

Prediction index: 14
Source row(s): 199
Candidate: 7.4.1.2
Context: <string>7.4.1.2</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.6901813179209421
Repeated prediction rows: 1

Prediction index: 15
Source row(s): 200
```

```
Candidate: 64.98.122.70
Context: <string>64.98.122.70</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.6306413358746117
Repeated prediction rows: 1

Prediction index: 17
Source row(s): 12, 39, 40, 42, 49, 136, 137, 138, 147, 202, 203, 214, 216, 223
Candidate: 12.6.0.1
Context: <string>12.6.0.1</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.8554291613663688
Repeated prediction rows: 5

Prediction index: 19
Source row(s): 180, 204, 205, 208, 209
Candidate: 257.0.1.38
Context: <string>257.0.1.38.116</string>
Actual label: negative
Model prediction: negative
```

```
Outcome: TN
Model score: 0.3569596547835475
Repeated prediction rows: 4

Prediction index: 21
Source row(s): 206
Candidate: 256.0.1.26
Context: <string>tincan_register_device_256.0.1.26.113</string>
Actual label: negative
Model prediction: positive
Outcome: FP
Model score: 0.6764016336122133
Repeated prediction rows: 1

Prediction index: 22
Source row(s): 207
Candidate: 257.0.1.38
Context: <string>tincan_register_device_257.0.1.38.116</string>
Actual label: negative
Model prediction: negative
Outcome: TN
Model score: 0.38467565709737667
Repeated prediction rows: 1
```

```
Prediction index: 25
Source row(s): 46, 210
Candidate: 135.0.0.22
Context: <string>135.0.0.22.118 (206072690)</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7828800320085034
Repeated prediction rows: 1

Prediction index: 27
Source row(s): 10, 134, 152, 212
Candidate: 3.6.1.4
Context: <string>3.6.1.4</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7770807151755506
Repeated prediction rows: 1

Prediction index: 32
Source row(s): 123, 150, 217
Candidate: 15.21.2.145
Context: <string>15.21.2.14594</string>
```

```
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7929243545627779
Repeated prediction rows: 1

Prediction index: 35
Source row(s): 220
Candidate: 5.12.20.4
Context: <string>5.12.20.4.10.18.5.5.2</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7321275029161977
Repeated prediction rows: 1

Prediction index: 36
Source row(s): 221
Candidate: 10.18.5.5
Context: <string>5.12.20.4.10.18.5.5.2</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.6509943259180028
```

```
Repeated prediction rows: 1

Prediction index: 42
Source row(s): 130, 132, 227, 229
Candidate: 255.255.255.255
Context: <string>255.255.255.255</string>
Actual label: positive
Model prediction: positive
Outcome: TP
Model score: 0.7309358316638832
Repeated prediction rows: 2

Prediction index: 43
Source row(s): 131, 133, 228, 230
Candidate: 0.0.0.0
Context: <string>0.0.0.0</string>
Actual label: negative
Model prediction: negative
Outcome: TN
Model score: 0.2648769552164866
Repeated prediction rows: 2

END OF INSPECTION
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ find . -maxdepth 3 -type f \( -name "*.py" -o -name "*.ipynb" \)
./phase_b_examples_backup.py
./phase_b_controlled_tests.py
./phase_b_evidence.py
./leakage_check.py
./octet_check.py
./phase_b_examples.py
./investigate.py
./lab01_pipeline.py
./phase_e_inspect.py
./lab01_phaseD_historical.py

┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ grep -Rn "random_state\|random.seed\|np.random.seed" . --include="*.py"
./lab01_pipeline.py:17:SEED = 42; np.random.seed(SEED)
./lab01_pipeline.py:75:              ("clf", LogisticRegression(max_iter=2000, class_weight="balanc
ed", random_state=SEED))])
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ sed -n '1,120p' lab01_pipeline.py
"""CIP-B107-LAB01 - offline IP-evidence triage pipeline (Phases A, B, C, E).
Run:  python lab01_pipeline.py <path_to_folder_with_supplied_files>
Outputs go to ./out/.  Reads originals only; never modifies them."""
import sys, os, json, hashlib, ipaddress, re, datetime as dt, platform
import pandas as pd, numpy as np, sklearn
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.dummy import DummyClassifier
from sklearn.metrics import (confusion_matrix, classification_report, accuracy_score,
                             precision_score, recall_score, f1_score, roc_auc_score,
                             average_precision_score)
import matplotlib; matplotlib.use("Agg"); import matplotlib.pyplot as plt

SRC = sys.argv[1] if len(sys.argv) > 1 else "."
OUT = "out"; os.makedirs(OUT, exist_ok=True)
SEED = 42; np.random.seed(SEED)
p = lambda f: os.path.join(SRC, f)

# ---------- Phase A: manifest ----------
tz_wat = dt.timezone(dt.timedelta(hours=1))
rows = []
for f in ["dataset_small.csv","dataset_train.jsonl","dataset_test.jsonl","result.csv","fine_tuning_s
```

```
teps.txt"]:
    b = open(p(f), "rb").read(); now = dt.datetime.now(dt.timezone.utc)
    rows.append(dict(filename=f, sha256=hashlib.sha256(b).hexdigest(), bytes=len(b),
                     timestamp_utc=now.isoformat(timespec="seconds"),
                     timestamp_wat=now.astimezone(tz_wat).isoformat(timespec="seconds"),
                     transfer_source="ICDFA supplied package", examiner="<YOUR NAME>"))
pd.DataFrame(rows).to_csv(f"{OUT}/evidence_manifest.csv", index=False)

df = pd.read_csv(p("dataset_small.csv"))
print("records", len(df), "| columns", list(df.columns))
print(df["class"].value_counts().to_dict(), "| nulls", int(df.isna().sum().sum()),
      "| duplicate rows", int(df.duplicated().sum()))
load = lambda f: pd.DataFrame([json.loads(l) for l in open(p(f)) if l.strip()])
tr, te = load("dataset_train.jsonl"), load("dataset_test.jsonl")
tr["y"] = (tr.completion.str.strip() == "positive").astype(int)
te["y"] = (te.completion.str.strip() == "positive").astype(int)
print("train", len(tr), tr.y.value_counts().to_dict(), "| test", len(te), te.y.value_counts().to_dic
t())

# ---------- Phase B: regex vs ipaddress vs source label ----------
RX = re.compile(r"\d{1,3}(?:\.\d{1,3}){3}")
def strict(s):
    try: ipaddress.IPv4Address(s); return True
    except Exception: return False
```

```
rev = []
for i, r in df.iterrows():
    m = RX.findall(r.text)
    rev.append(dict(source_row=i+2, context=r.text.strip(), candidate=r.ip,
                    regex_match=r.ip in m, ipv4_valid=strict(str(r.ip)),
                    source_label="positive" if r["class"] ≠ "N" else "negative", analyst_note=""))
rev = pd.DataFrame(rev)
def note(r):
    if r.ipv4_valid and r.source_label == "negative": return "Strictly valid IPv4 but source says ne
gative - context/version-like string; review"
    if (not r.ipv4_valid) and r.source_label == "positive": return "Source positive but fails strict
 IPv4 (e.g. leading zeros/octet) - preserve disagreement"
    if not r.regex_match: return "Candidate not matched by simple regex in context"
    return ""
rev["analyst_note"] = rev.apply(note, axis=1)
rev.to_csv(f"{OUT}/candidate_review.csv", index=False)
print(pd.crosstab(rev.source_label, rev.ipv4_valid, rownames=["source"], colnames=["strict_valid"]))

# ———————— Phase C: leakage + models ————————
overlap = te.prompt.isin(set(tr.prompt)).sum()
print(f"LEAKAGE: {overlap}/{len(te)} test prompts are exact duplicates of training prompts")

def rule(prompt):
    ctx, cand = prompt.rsplit(";", 1); cand = cand.replace("→", "").strip()
```

```
    if not strict(cand): return 0
    # candidate must be a standalone token, not glued to digits/letters/dots
    pat = r"(?<![\w.])" + re.escape(cand) + r"(?![\w.])"
    return int(re.search(pat, ctx) is not None)
te["rule_pred"] = te.prompt.map(rule)

ml = Pipeline([("tfidf", TfidfVectorizer(analyzer="char", ngram_range=(1,4), lowercase=False)),
               ("clf", LogisticRegression(max_iter=2000, class_weight="balanced", random_state=SEED)
)])
ml.fit(tr.prompt, tr.y)
te["ml_pred"] = ml.predict(te.prompt); te["ml_score"] = ml.predict_proba(te.prompt)[:,1]
maj = DummyClassifier(strategy="most_frequent").fit(tr.prompt, tr.y); te["majority_pred"] = maj.pred
ict(te.prompt)

# ———————— Phase E: evaluation ————————
def cell(a, b): return {(1,1):"TP",(0,0):"TN",(0,1):"FP",(1,0):"FN"}[(a,b)]
te["actual"] = te.y.map({1:"positive",0:"negative"}); te["ml_pred_label"] = te.ml_pred.map({1:"posit
ive",0:"negative"})
te["outcome"] = [cell(a,b) for a,b in zip(te.y, te.ml_pred)]
te[["prompt","actual","ml_pred_label","ml_score","outcome","rule_pred","majority_pred"]].to_csv(f"{O
UT}/predictions.csv", index=False)
res = []
for name, col in [("majority","majority_pred"),("rule (regex+ipaddress)","rule_pred"),("TF-IDF char
+ LogReg","ml_pred")]:
```

```
    cm = confusion_matrix(te.y, te[col], labels=[0,1]); tn, fp, fn, tp = cm.ravel()
    res.append(dict(model=name, TP=tp, FP=fp, FN=fn, TN=tn, accuracy=accuracy_score(te.y, te[col]),
        precision=precision_score(te.y, te[col], zero_division=0), recall=recall_score(te.y, te[col]
),
        f1=f1_score(te.y, te[col], zero_division=0)))
res = pd.DataFrame(res).round(4); res.to_csv(f"{OUT}/metrics_summary.csv", index=False); print(res.t
o_string(index=False))
print("ML AUROC %.4f | AUPRC %.4f" % (roc_auc_score(te.y, te.ml_score), average_precision_score(te.y
, te.ml_score)))
print(classification_report(te.y, te.ml_pred, target_names=["negative","positive"], zero_division=0)
)

fig, ax = plt.subplots(1, 2, figsize=(9,3.8))
for a, (t, c) in zip(ax, [("TF-IDF + LogReg","ml_pred"),("Rule baseline","rule_pred")]):
    cm = confusion_matrix(te.y, te[c], labels=[1,0]); a.imshow(cm, cmap="Blues"); a.set_title(t)
    a.set_xticks([0,1], ["positive","negative"]); a.set_yticks([0,1], ["positive","negative"])
    a.set_xlabel("Predicted"); a.set_ylabel("Actual")
    for (i,j),v in np.ndenumerate(cm): a.text(j,i,v,ha="center",va="center",fontsize=14)
plt.tight_layout(); plt.savefig(f"{OUT}/confusion_matrices.png", dpi=150)
open(f"{OUT}/environment.txt","w").write(f"python {platform.python_version()}\npandas {pd.__version_
_}\nnumpy {np.__version__}\nscikit-learn {sklearn.__version__}\nseed {SEED}\n")
```

```
┌──(.venv)─(kali㉿kali)-[~/CIP-B107-LAB01/work]
└─$ python3 lab01_pipeline.py ~/CIP-B107_original/code
records 229 | columns ['ip', 'class', 'text']
{'A': 177, 'N': 32, 'B': 12, 'C': 4, 'E': 4} | nulls 0 | duplicate rows 77
train 183 {1: 163, 0: 20} | test 46 {1: 34, 0: 12}
strict_valid  False  True
source
negative         24      8
positive          4    193
LEAKAGE: 30/46 test prompts are exact duplicates of training prompts
                          model  TP  FP  FN  TN  accuracy  precision  recall      f1
                       majority  34  12   0   0    0.7391     0.7391  1.0000 0.8500
rule (regex+ipaddress)  21   2  13  10    0.6739     0.9130  0.6176 0.7368
  TF-IDF char + LogReg  34   1   0  11    0.9783     0.9714  1.0000 0.9855
ML AUROC 0.9902 | AUPRC 0.9965
              precision    recall  f1-score   support

    negative       1.00      0.92      0.96        12
    positive       0.97      1.00      0.99        34

    accuracy                           0.98        46
   macro avg       0.99      0.96      0.97        46
weighted avg       0.98      0.98      0.98        46
```